Etiology and Pathogenesis

Rubella is an enveloped RNA virus in the Togaviridae family. The virus is spread through direct or droplet contact from nasopharyngeal secretions. Infected individuals shed virus for 5 to 7 days before and up to 14 days after the onset of disease, with viremia unlikely after the rash appears. In most individuals, infection leads to lifelong immunity.

Congenital rubella occurs when a nonimmunized, susceptible pregnant woman is exposed to the virus. Transplacental infection of the fetus occurs during the viremic stage. The risk is greatest when fetal exposure happens in the first trimester. Congenitally infected infants may shed the virus through urine, blood, and nasopharyngeal secretions for up to 12 months after birth, making them a potential source of infection for other susceptible individuals.

Clinical Findings

History

Primary rubella infection is typically a mild, subclinical disease, particularly in adults. The prodrome includes low-grade fever, myalgias, headache, conjunctivitis, rhinitis, cough, sore throat, and lymphadenopathy. These symptoms may last up to 4 days and often resolve as the rash appears. Up to 50% of children with primary rubella infection may have a subclinical course or present with lymphadenopathy or rash alone (without prodrome). In contrast, older adults may experience more severe and prolonged prodromal symptoms, which can make clinical differentiation from rubeola (measles) difficult. The presence of Koplik spots in the oral mucosa favors rubeola.

As the prodrome resolves and the rash develops, some patients may exhibit an enanthem consisting of tiny red macules on the soft palate and uvula (Forschheimer spots). This finding is not diagnostic for rubella.

Cutaneous Lesions

The exanthem appears 14 to 17 days after exposure and consists of pruritic pink to red macules and papules that begin on the face and rapidly spread to the neck, trunk, and extremities. Lesions on the trunk may coalesce, while those on the extremities tend to remain discrete. The rash typically lasts 2 to 3 days, in contrast to rubeola, which persists longer and clears starting from the head and neck. Desquamation may occur after the rash resolves.

Related Physical Findings

Lymphadenopathy is most prominent in the posterior cervical, suboccipital, and postauricular lymph nodes and may be noted up to 7 days before rash onset. Node enlargement can persist for several weeks.

Arthritis occurs in up to 70% of infected adults, particularly women. Both small and large joints may be involved. Joint symptoms often begin as the rash fades and can last for several weeks. In some cases, symptoms may become persistent or recurrent, and joint swelling may progress to effusion.

Laboratory Tests

A complete blood cell count typically shows leukopenia with relative neutropenia.